

Evaluating No-Code Agentic AI Frameworks for Enterprise Automation: A Practical Comparison of Make.com and n8n.io

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Abstract—The rapid adoption of Agentic Artificial Intelligence across enterprises has created demand for accessible automation platforms that do not require traditional programming expertise. No-code workflow tools such as Make.com and n8n.io have emerged as practical solutions; however, peer-reviewed literature directly comparing these platforms for Agentic AI automation remains scarce. This paper addresses that gap through a practitioner-led comparative study. Three functionally identical Agentic AI workflows — an AI travel planning assistant, an automated stock market alert system, and an AI search quality evaluator — were independently implemented on both Make.com and n8n.io and evaluated across seven criteria. Findings indicate that n8n.io demonstrated superior execution speed, fewer build-time errors, higher AI accuracy, stronger debugging feedback, and greater cost efficiency through its credit-free model, while Make.com offered advantages in public workflow sharing and beginner accessibility. The study concludes that platform selection should be guided by specific project requirements rather than a single universal recommendation, and identifies the absence of native fine-tuning and Retrieval-Augmented Generation support as a shared limitation of both platforms.

Index Terms—Agentic AI, No-Code Automation, Make.com, n8n, Enterprise Automation, Workflow Orchestration, EU AI Act, Prompt Engineering

I. INTRODUCTION

Agentic Artificial Intelligence (AI) refers to autonomous systems capable of reasoning, planning, and executing multi-step tasks independently through the use of tools, memory, and real-time data. As enterprises increasingly adopt such capabilities, no-code automation platforms have emerged as accessible solutions for building AI-driven workflows without traditional programming expertise.

Among the available no-code platforms, Make.com and n8n.io represent two of the most prominent options. This research began with a practical interest in building Agentic AI workflows using no-code tools. After independently constructing projects on both Make.com and n8n.io, clear differences in performance, usability, cost, and reliability became apparent. These observed differences motivated a structured comparison between the two platforms.

A review of existing literature revealed that very few academic papers compare no-code automation tools, and almost none provide a direct practitioner-led comparison of

Make.com and n8n.io for Agentic AI automation. This scarcity of research presented an opportunity to contribute an original comparative study.

The purpose of this paper is to determine which platform offers a smoother, more efficient, and more enterprise-suitable experience for building Agentic AI workflows. The research is guided by three questions:

- RQ1: How do Make.com and n8n.io compare in performance and usability for Agentic AI workflows?
- RQ2: Which platform offers greater efficiency, reliability, and cost-effectiveness?
- RQ3: What limitations do these no-code platforms exhibit for enterprise-scale AI needs?

II. LITERATURE REVIEW

The academic treatment of no-code workflow automation platforms remains limited, although a small number of relevant works exist. A study titled “Optimizing the Automation Process with n8n” examines the architecture of modern visual workflow tools, contrasting the data-handling methodology of n8n against platforms such as Zapier and Make.com. That work focuses heavily on self-hosted deployment via Docker and on workflow execution efficiency, offering valuable architectural insight but limited evaluation of Agentic AI capabilities specifically.

In addition, an industry-compiled research dataset titled “N8N vs Make.com vs Zapier: Which Automation Tool is Best Overall” explores scalability, AI capabilities, and data limitations using real-world workflow data drawn from more than 120 expert contributors. While this provides useful breadth, it represents aggregated industry opinion rather than a controlled, practitioner-led implementation study.

Beyond these contributions, peer-reviewed literature directly comparing Make.com and n8n for Agentic AI automation is notably scarce. A likely explanation is the pace of change within the no-code and AI automation sector; these platforms evolve so rapidly that traditional peer-reviewed academic publishing struggles to keep pace with their development.

Furthermore, the regulatory dimension introduced by the European Union Artificial Intelligence Act has received mini-

mal attention in the context of no-code automation tooling. As enterprises within the European Union become subject to risk-based AI compliance obligations, the privacy and governance characteristics of these platforms acquire increased significance.

This paper addresses these combined gaps by providing a structured, multi-criteria comparison grounded in the direct implementation of identical workflows on both platforms, with explicit attention to enterprise readiness and regulatory considerations.

III. METHODOLOGY

A practitioner-led experimental methodology was adopted for this study. Three functionally identical Agentic AI workflows were designed and independently implemented on both Make.com and n8n.io, ensuring functional equivalence across platforms. This controlled approach enables direct comparison while holding workflow purpose and complexity constant.

A. Test Case Workflows

Test Case 1 — AI Travel Planning Assistant: A workflow that receives user travel preferences via webhook, processes them through a large language model, and delivers a personalised day-by-day itinerary via email.

Test Case 2 — Automated Stock Market Alert System: A workflow that monitors fifteen equities against user-defined target prices, applies conditional logic, and dispatches email alerts only when target conditions are satisfied.

Test Case 3 — AI Search Quality Evaluator: A workflow that reads search queries and result snippets from a spreadsheet, classifies their relevance using an AI model, and writes structured evaluations back to the data source.

B. Evaluation Criteria

Both platforms were evaluated across seven criteria: technical capability, cost and scalability, usability and learning curve, error handling and fault tolerance, sharing and collaboration, AI model integration, and enterprise readiness.

C. Tools and Environment

On Make.com, workflows were built using sequential visual modules and the platform’s native AI Toolkit. On n8n.io, equivalent workflows were constructed using interconnected nodes, with the Groq Chat Model (llama-3.3-70b-versatile) providing AI inference, selected for its no-cost availability and processing speed. Webhook-based workflows were triggered using Hoppscotch for testing. Google Sheets served as the data source for the stock alert and search quality evaluation workflows, with Gmail providing email delivery.

IV. IMPLEMENTATION

A. Initial Experience

As a first-time user of no-code automation, the initial experience on Make.com involved a significant learning curve. Core concepts such as webhooks and triggers were initially unfamiliar. However, the visual nature of the platform proved intuitive:

observing each node trigger sequentially, and progress to the next node only upon successful execution, provided clear feedback on workflow behaviour.

Webhook-based workflows were triggered using Hoppscotch, which sends a request to the unique HTTPS webhook URL, initiating the workflow. Upon successful execution of all nodes, an email was generated and received. The generation of this first automated email alert represented a significant milestone, demonstrating that a no-code platform could produce professional, automated communication suitable for enterprise marketing and alerting purposes.

B. Conditional Logic Challenges

A notable challenge arose during implementation of the Stock Market Alert System. The conditional (IF) node initially produced inverted results — generating six alerts where only four were expected. This was diagnosed as a configuration issue in the conditional logic, compounded by incorrect node placement. Once the logic was corrected and the IF node was positioned appropriately before the email node, the system produced accurate results. Importantly, this challenge was identified as a learning-curve issue rather than a limitation of the platform itself.

Subsequent testing, performed by altering values within the data source, confirmed that the system reliably dispatched alerts only when target conditions were genuinely met.

C. Cross-Platform Observations

When the same workflows were rebuilt on n8n.io, several differences became apparent. The platform presented a distinct dark-themed interface and, critically, operated without a credit-based usage system. This credit-free model permitted unlimited testing, in contrast to Make.com, where finite monthly credits encouraged conservation rather than free experimentation.

In terms of performance, n8n.io executed workflows marginally faster, with alerts received a few seconds sooner, and exhibited no perceptible interface lag. The most significant observed difference concerned iterative development feedback: on n8n.io, each modification to a prompt produced a visible change in output — even if not yet the desired result — providing valuable diagnostic feedback for debugging. On Make.com, by contrast, output changes were often not visible until the prompt was fully corrected, making incremental debugging more difficult.

A further architectural observation concerned the use of AI for deterministic tasks. Where the language model was assigned numerical comparison, inconsistencies arose. Relocating deterministic comparison logic to dedicated conditional nodes improved reliability, illustrating the principle of selective AI invocation: reserving AI for tasks genuinely requiring natural language understanding, while delegating deterministic operations to rule-based logic.

V. RESULTS AND ANALYSIS

The comparative evaluation across all seven criteria is summarised in Table I. Findings are derived from direct implementation experience across the three test cases.

TABLE I
COMPARATIVE EVALUATION OF MAKE.COM AND N8N.IO

Criterion	Make.com	n8n.io
Execution Speed	Slower	Faster
Build Errors	More frequent	Fewer
Usability	Visually intuitive	Easier in practice
Cost Model	1000 free ops/month	Credit-free testing
Error Handling	Basic	Advanced
Public Sharing	Public URL	No public sharing
AI Flexibility	Make AI Toolkit	Multiple models
Data Privacy	Limited control	Greater control

A. Performance and Reliability

In direct implementation, n8n.io demonstrated faster execution and fewer build-time errors across the test cases. The AI output on n8n.io was observed to be more accurate, and the platform’s capacity to display incremental output changes during prompt refinement materially assisted the debugging process.

B. Cost and Testing Freedom

A significant practical finding concerned the cost model. The credit-free nature of n8n.io removed the psychological constraint experienced on Make.com, where finite credits encouraged users to conserve rather than test freely. This testing freedom proved particularly valuable during the iterative debugging of variable mapping issues, which occurred on both platforms but incurred credit cost only on Make.com.

C. Learning Transfer

An additional observation concerned learning transfer. Prior experience gained on Make.com enabled faster resolution of equivalent issues on n8n.io. For instance, variable mapping problems were resolved efficiently on n8n.io by executing previous nodes — a technique informed by earlier Make.com experience.

D. Sharing and Privacy

A clear divergence emerged in sharing capability. Make.com supports public workflow sharing via URL, advantageous for students and practitioners demonstrating their work. By contrast, n8n.io restricts sharing to specified users or JSON export. While this limits public visibility, it offers stronger data privacy — a consideration aligned with European Union AI Act requirements, which mandate that while AI behaviour should be transparent, the underlying process and user data should remain protected.

VI. DISCUSSION

The results indicate that no single platform is universally superior; each demonstrates distinct strengths. n8n.io proved better suited to contexts prioritising execution performance, reliability, cost efficiency, AI model flexibility, and data privacy. Make.com offered advantages in beginner accessibility through its intuitive visual interface and in public workflow sharing.

A noteworthy and somewhat counterintuitive finding concerns beginner onboarding. Although Make.com is frequently positioned as the more beginner-friendly platform, this study suggests that beginners may benefit from starting with n8n.io. Its credit-free model permits unlimited experimentation without the anxiety of depleting finite resources, allowing new users to learn through repeated testing. Practitioners may subsequently transition to Make.com, by which point they possess sufficient competence to operate effectively within its credit constraints.

VII. LIMITATIONS AND FUTURE WORK

This study reflects the experience of a single practitioner across three test cases; broader multi-practitioner studies would strengthen generalisability. Performance observations were qualitative rather than derived from instrumented benchmarking, representing a clear direction for future quantitative research.

Both platforms exhibited a shared limitation: neither provides native support for model fine-tuning or Retrieval-Augmented Generation (RAG). Workflows relying solely on prompt engineering therefore encounter scalability constraints, including context-window limitations when processing large datasets. This was particularly evident in the search quality evaluation workflow, which functioned effectively for a small dataset but would not scale to enterprise volumes without RAG integration and fine-tuning informed by Reinforcement Learning from Human Feedback (RLHF).

VIII. CONCLUSION

This paper presented an original, practitioner-led comparison of Make.com and n8n.io for Agentic AI enterprise automation, grounded in the implementation of three functionally identical workflows on each platform. The findings demonstrate that n8n.io offers advantages in execution speed, reliability, cost efficiency, and data privacy, while Make.com excels in public sharing and beginner accessibility.

The central conclusion is that platform selection must be guided by the specific needs of the project. This principle may be understood through a metaphor: the landscape of no-code AI tools resembles an ocean filled with fish of varying sizes, where the AI Project Manager is the fisherman and the boat represents the scale and requirements of the project. A small boat is suited to catching smaller fish, while a larger boat is required for larger ones. Likewise, each automation platform is best suited to projects within a particular range of complexity and scale. The essential competency of the AI Project Manager, therefore, is not mastery of a single tool,

but the judgement to select the most appropriate tool for the specific demands of each project.

As Agentic AI adoption accelerates and regulatory frameworks such as the EU AI Act mature, this capacity for informed platform selection will become an increasingly valuable competency for AI Project Managers and enterprise decision-makers alike.

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